

Anti-Jamming techniques for aviation GNSS-based navigation systems: Survey

SENHAJI.MOUHADDIB ISSAM
ENSA, Ibn Tofail University
Kenitra, Morocco
issam.senhaji-mouhaddib@uit.ac.ma

ADDAIM ADNANE
EMI, Mohammed V University
Rabat, Morocco
adnane.addaim@um5.ac.ma

AIT MADI ABDESSALAM
ENSA, Ibn Tofail University,
Kenitra, Morocco
abdessalam.aitmadi@uit.ac.ma

Abstract— Aviation is now heavily dependent on GNSS signals to get the PNT (Position, Navigation, Timing) elements with a high degree of accuracy, reliability and continuity. The International Civil Aviation Organization (ICAO) with the support of regional organizations like the FAA (Federal Aviation Authority) and Eurocontrol are now moving to concretize the recommendations of GANP (Global Air Navigation Plan) which make from GNSS the key technology of CNS/ATM concept (Communication, Navigation, Surveillance/Air Traffic Management). So, it is obvious that the security and the integrity of GNSS signals must be a concern of all actors in military and civil aviation. In this paper, we will present a survey of the anti-jamming techniques for airborne GNSS-based navigation systems and some approaches to detect, mitigate the jamming and provide a secure GNSS-based system.

Keywords—GNSS, Jamming, Spoofing, Navigation, Aeronautics

I. INTRODUCTION

In aviation, GNSS is used in all flight phases and intended to ensure a good navigation performance in comparison with other navigation systems like the INS (Inertial Navigation System) that has some limitations due to cumulated errors related with the duration of flight.

Area Navigation (RNAV) is based on GNSS signals, particularly in landing phase that requires a very accurate RNP (Required Navigation Performance) and especially in fields not equipped with Radio NavAids (ILS, DME, VOR...).

Also, the Flight Management System and almost all ATM systems and operations (Automatic Dependent Surveillance B/C) need timing synchronization from GNSS signals to guarantee a full secure integrated aviation system.

Besides that, the guidance of some weapons carried on board military aircrafts, is ensured by GPS signals to a coordinate-defined point.

However, the estimated power of GNSS signals received on earth is approximatively -130 dBm, and are dominated by white Gaussian noise. This results in the fact that GNSS signals are very weak and susceptible to all types of interferences: intentional and unintentional.

There are several types of threat that can interfere intentionally with a GNSS receiver's ability to receive and process GNSS signals. The table 1 below summarizes different types of intentional interferences.

Threat type	description	Impact on system
jamming	any radio frequency interference signals that deteriorate GNSS reception and accuracy	Loss of signal or ranging errors

spoofing	attempts to deceive a GPS receiver by broadcasting a slightly more powerful signal than that received from the GPS satellites, structured to resemble a set of normal GPS signals causes the receiver to determine its position to be somewhere other than where it actually is.	False location and time readings
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Table 1: Types of intentional interferences

In recent years, several incidents were reported mentioning the impact of jamming and spoofing on GNSS signals. In fact, more than 250 incidents of GPS disruption have been reported by pilots through NASA's Aviation Safety Reporting System since 2013, and a total of 815 incidents reported to Eurocontrol so far in 2018 [1]. In addition to that, the following incidents are the more attractive ones:

- American UAV (RQ-170) captured by Iran using a GPS spoofer in 2011;
- GBAS jammed by a PPD in Newark Airport in USA resulting on the unavailability of GNSS-based approaches;
- 1000 airplanes in the South-Korean Airspace suffered from jamming done by North-Korean military systems...etc..

Firstly, and before starting the analysis of the situations related to jamming GNSS signals in the aviation model, we have to list almost all the impacts of such intentional interferences on the performances of navigation and surveillance functions of a CNS/ATM system.

The loss of GNSS signals may have a heavy losses in lives and infrastructures. In each flight phase, and in each airspace throughout the world, it is necessary to have a specific RNP performance which is more critical to respect in the final approach segment. The table 2 below summarizes some of them [2].

Type of Operation	HOR/VERT Accuracy (95%)	HAL/VAL
Enroute Oceanic	3700 m / NA	7400 m / NA
Enroute continental	3700 m / NA	3704 m / NA
Enroute Terminal	740 m / NA	1852 m / NA
APV-I	16 m / 20 m	40 m / 50 m
APV-II	16 m / 8 m	40 m / 20 m
Category I	16 m / 4 m	40 m / 10 m
Category II	6,9 m / 2 m	17,3 m / 5,3 m
Category III	6,9 m / 2 m	15,5 m / 5,3 m

Table 2: RNP related to flights phases and airspaces

In the same context, the loss of GNSS signals may be critical to the safety of flight. As an example, losing the EGPWS function (Enhanced Ground Proximity Warning System) which takes position input from GPS can lead the airplane crew, in some weather and night conditions, to take a bad decisions. The following table lists some of impacts on navigation and surveillance systems generated by the loss of GNSS [3]:

	Impact on system	Effects on cockpit		
		A300	A220	A320
NAVIGATION	Downgraded aircraft position computation	NAV GPS Xfault GPS PRIMARY LOST	GNSS NOT AVAIL	NAV GPS Xfault GPS PRIMARY LOST
	Impact on RNP OPS		UNABLE RNP	
	Loss of RNAV capability		APPROACH NOT AVAIL	
	Loss of RNP AR capability	GPS PRIMARY LOST		
SURVEILLANCE	Loss of predictive TAWS	TERR MODE FAULT	TAWS MAP FAIL	NAV GPWS TERR DET FAULT
	Loss of ADS-B Out reporting	N/A	ADS B OUT FAIL	NAV ADS B FAULT
	Loss of ADS-B In function	N/A	N/A	

Table 3: Loss of GNSS impacts on navigation and surveillance functions

Finally, in this paper, the investigation is more oriented to jamming analysis than spoofing detection and mitigation techniques.

II. RELATED WORK

Generally, jammers are classified into three broad categories: Narrowband Jammers (NBJ), Spread Spectrum Jammers (SSJ), and Wideband Gaussian Jammers (WGJ).

Typical jamming signals can be classified into four classes:

- Class I: continuous wave signal;
- Class II: chirp signal with one saw-tooth function;
- Class III: chirp signal with multi saw-tooth functions;
- Class IV: chirp signal with frequency bursts.

In the literature, several studies were conducted to find software and hardware solutions to mitigate GNSS jamming. We can say, by Electronic Warfare terminology, that these solutions are named GNSS Electronic Protection Measures (EPM).

The GNSS EPM are carried out in time, frequency and spatial domains and are implemented in various parts of GNSS receivers (see Fig.1): antenna, front-end, pre and post-correlation processing and finally the positioning domain. It is worth to mention that some innovative studies have proposed a multi-domain solution like the joint frequency-time domain mitigation technique.

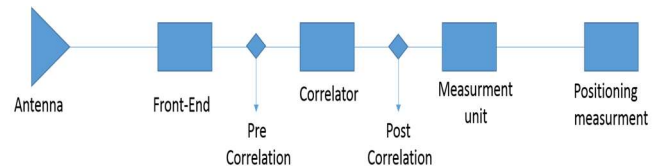


Fig. 1. Various parts of GNSS receiver implementations

As a summary, the table 2 contains some GNSS EPM gathered from referred documents:

TABLE 4 : GNSS EPM SUMMARY

Level	EPM Title	Details
Antenna	CRPA: Controlled Reception Pattern Antenna	Detection and mitigation in the spatial domain. Usually used in military receivers and GBAS systems.
	Anti-jamming null-steering control technique in dynamic scenes [4]	Improve the anti-jamming performance of the spatial filtering in dynamic scenes (Motion) by deepening the depth of the null
	Dual polarized GPS antenna array [5]	Can mitigate twice the number of jamming signals more than a conventional antenna
	Innovative helical antenna for aerial platforms [6]	This type of antenna has a lower side lobes which are more susceptible to jamming
	Two stage interference suppression scheme [7]	A technique implemented before the despreading of GNSS signals and dealing with both jamming and spoofing.
Front-End	AGC monitoring	Variations in gain are used for jamming detection
	ADC test samples	Analog to Digital raw samples are distributed normally to detect jamming. They are also used for frequency-time detection methods
Pre-correlation	Filtering	Notch filter (Time domain) adaptive filter (Frequency domain)
	Pulse blanking [8]	Jamming detection with power exceeding a certain threshold and shorter than a certain time period
	Null steering	Suppressing undesired signals in the direction of appearance
	DoI	Measuring the direction of jamming
Post-correlation	Beamforming	Increasing the signal strength in the direction of interest
	DOA	Measuring Direction of Arrival of signal of interest
Positioning	ABIA : Avionics-Based GNSS Integrity Augmentation [9]	An extended Kalman Filter is used for multi-sensor data fusion of different navigation sensors to obtain PVA (Position, Velocity, and Attitude) elements.
	Improved Fuzzy Weighted Least Square Method in combined GPS/GLONASS [10]	WLS makes the sum of the squares of the errors as small as possible to find the best estimation of PNT elements.

Moreover, a few studies have focused on the dynamic changes related to the motion of GNSS receivers (ex: aircraft, missile...). In the other hand, some experiments were conducted on the flight and demonstrated that a low power of jammer (200 mW) was sufficient to cause unreliable position information in the Flight Management System of an aircraft [11].

We may conclude, in regards to what was said previously, that each technique used to detect and mitigate GNSS jamming has its own advantages and disadvantages, both in processing and cost. Also, almost all simulations were not performed against all types of jamming signals generated by PPD or frequency generated lab-based.

In addition to that, and with studying some published documents [12] [13] [14] [15] [16], it has been shown that location of jamming source has not been really initiated in the case of an airplane or infrastructure of an airfield.

Consequently, a global study must be done to elaborate a strategy that could be able to detect, mitigate and locate GNSS jamming in a dynamic environment under NAVWAR (Navigation Warfare) conditions.

III. ONGOING RESEARCH

Providing a secure and immune GNSS-based system against all sort of jamming is crucial to maintain and guarantee the full services and functions of CNS/ATM and thus permit a safe landing, ensure take-off within time limits, and save money to all operators and companies due to possible delays [17].

In our project, we are aiming to begin our investigation in recent studies dealing with jamming GNSS signals. In our best knowledge, a few documents have studied the case of jamming GNSS receivers of aircrafts and the infrastructures of airports (GBAS...). The only one who was really involved in this case is Roberto Sabatini [2] [9] who worked on a system able to detect, mitigate and elaborate an avoidance volume to escape from the jamming envelope (ABIA: Avionics-Based GNSS Integrity Augmentation). However, and by lack of information, it is not proven yet if this system has accomplished the mission.

From our perspective, we are projecting to find a whole solution to mitigate jamming where all systems are involved: Jammer, aircraft, satellite, GBAS...

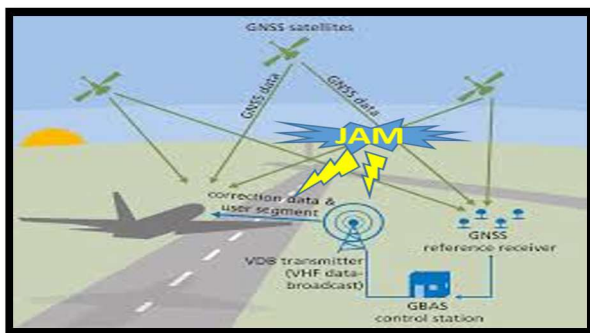


Fig. 2. Scenario of the project

To do so, simulations will be done on MATLAB/SIMULINK and probably other tools based on the next scenario: jammer will be held in the threshold of the runway against an aircraft in the final approach of an airport equipped with a GBAS system.

The objective of such simulation is to validate our EPM against all types of Jamming signals, and the most important thing is to locate the source of the interference to let the ATM services intervene in the fastest way. To determine the Direction Of Arrival of the emitting source two techniques will be tested: the amplitude comparison and the principle of interferometry. However, and to find the exact position of the jammer, triangulation technique is not suitable to a small area related to our scenario. In our simulations, TDOA (Time Differential Of Arrival) may be the best technique for the location of the Electronic Attack device.

Also, we will be working to find a backup solution to GNSS as a navigation tool, like the amelioration of INS (Inertial Navigation System) to reduce position errors, and RAIM (Receiver Autonomous Integrity Monitoring) to support the crew of the aircraft to be alerted when jamming occurs and consequently track a re-routing as an avoidance tactic.

IV. CONCLUSION AND PERSPECTIVES

Electronic Warfare concerns basically three components: Electronic Support, Electronic Attack and Electronic Protection.

In this paper, we are dealing with EA and EP of a GNSS based signals environment. It is for sure not common to talk about jamming airborne GNSS receivers or ground based augmentation systems in the civil aviation, but recent incidents throughout the world have incited ICAO, FAA and Eurocontrol to support researches in this field and to train pilots and air controllers to be accommodated to deal with this type of phenomena, initially by generating NOTAMS (Notice To Air Men) in case of military exercises in which jamming GPS or GALILEO is scheduled.

Providing a global solution to this type of problem, will help the CNS/ATM system to work under safe conditions and thus economize cost and save lives of passengers.

Finally, the recent studies about this topic show a major concern to GNSS jamming by technical means with enhancing the performance of GNSS receivers by filtering or conceiving a new antennas able to reduce the side lobes pattern. In our context, a study must be performed about the feasibility of such antenna to be implemented on the fuselage of an aircraft with respect to aerodynamics principles and electromagnetic compatibility with other avionics systems.

Future work will be conducted on schematizing the whole system englobing jammer and victims (Aircraft, GBAS, ADS B, Satellite) with all properties related to jamming signals types and the subdivisions of GNSS: GPS, GALILEO, BEIDOU, and GLONASS.

REFERENCES

- [1] GNSS interferences and commercial aviation – real world impacts. Guy Buesnel, Steve Hickling November 2018.
- [2] Global navigation satellite systems performance analysis and augmentation strategies in aviation. Roberto Sabatini, Terry Moore, Subramanian Ramasamy 2017.
- [3] GNSS vulnerabilities. MIDANPIRG/17 RASG MID WP/9 2019.
- [4] An anti jamming null steering control technique based on double projection in dynamic scenes for GNSS receivers. Hao Wang, Qing Chang, Yong Xu April 2019.
- [5] Dual-polarized GPS antenna array algorithm to adaptively mitigate a large number of interference signals. Kwansik Park, Dongkook Lee, Jiwon Seo 2018.
- [6] Design of an innovative helical antenna for aerial platforms to enable jam resistant reception of GNSS signals. Inamullah, Talha Farooq Hashmi, Irfan Majid, April 2019.
- [7] A Two-Stage Interference Suppression Scheme Based on Antenna Array for GNSS Jamming and Spoofing. Jiaqi Zhang, Xiaowei Cui, Hailong Xu and Mingquan Lu September 2019.
- [8] GNSS jamming: a short survival guide. Daniele Borio 15 November 2016.
- [9] Avionics-Based GNSS Integrity Augmentation Performance in a Jamming Environment. Roberto Sabatini, Terry Moore, Subramanian Ramasamy 2015.
- [10] Jamming Mitigation using an Improved Fuzzy Weighted Least Square Method in Combined GPS and GLONASS Receiver. Mohammad-Reza Mosavi, Atieh Khavari, Amir Tabatabaei, Mohammad-Javad Rezaei 2017.
- [11] Jamming of aviation GPS receivers: investigation of field trials performed with civil and military aircraft. Pascal Truffer, Maurizio Scaramuzza, Marc Troller, skyguide, Swiss Air Navigation Services Ltd. Marc Bertschi, Swiss Air Force 2017.
- [12] Intelligent GNSS/INS integrated navigation system for a commercial UAV flight control system. Guohao Zhang, Li-Ta Hsu* July 2018
- [13] Jamming Efficacy of Variable Altitude GPS Jammer against Airborne GPS Receiver, Theoretical Study and Parametric Simulation. Ahmad Esmailkhah February 2018
- [14] Measurement of GNSS signal interference by a flight laboratory. Andrej Nováka*, Karel Havela, Martin Bugaja 2018
- [15] Multi-Radio Integrated Navigation System M&S Software Design for GNSS Backup under Navigation Warfare. Heyone Kim 1, Junhak Lee 2, Sang Heon Oh 3, Hyounghmin So 4 and Dong-Hwan Hwang 1 February 2019
- [16] GNSS Navigation Threats Management on-Board of Aircraft . Elena Simona LOHAN*, 1, Ruben Morales FERRE1, Philipp RICHTER1, Emanuela FALLETTI2, Gianluca FALCO2, Alberto DE LA FUENTE3. September 2019
- [17] Operational Impacts of Intentional GPS Interference Approved by the Tactical Operations Committee March 2018